

# Lecture 24

## Model comparison part 2

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## Recap: Model selection

- ▶ So far we have many potential models in our toolbox.
- ▶ For a given dataset, how to determine whether a simple model is sufficient or if we need a very complex one?
- ▶ There is probably no “right” model in practice.
- ▶ A **statistical model** is just a mathematical representation of the system that includes errors and biases in the observational process.
- ▶ All models are basically some simplifications of the reality and the reality is never that simple.
- ▶ We want a model that is as simple as possible yet seems to explain the patterns in the data reasonably well.

## Outline of Chapter 5

- ▶ For model comparisons, there are a finite number of candidate models and we want to select one using
  - ▶ Bayes factors
  - ▶ Stochastic search variable selection
  - ▶ Cross validation
  - ▶ Deviance information criteria (DIC)
  - ▶ Watanabe-Akaike information criteria (WAIC)
- ▶ In cases where multiple models fit well, we will discuss
  - ▶ Bayesian model averaging (BMA)
- ▶ After selecting a model, we want to test whether it fits the data well
  - ▶ Posterior predictive checks

## Recap: Bayes factors (BF)

- ▶ The BF for model  $\mathcal{M}_2$  compared to model  $\mathcal{M}_1$  is

$$BF_{2,1} = \frac{\text{Posterior odds}}{\text{Prior odds}} = \frac{p(\mathcal{M}_2|\mathbf{Y})/p(\mathcal{M}_1|\mathbf{Y})}{p(\mathcal{M}_2)/p(\mathcal{M}_1)} = \frac{p(\mathbf{Y}|\mathcal{M}_2)}{p(\mathbf{Y}|\mathcal{M}_1)}.$$

- ▶ Rule of thumb:  $BF_{2,1} > 10$  is strong evidence for  $\mathcal{M}_2$ .
- ▶ Rule of thumb:  $BF_{2,1} > 100$  is decisive evidence for  $\mathcal{M}_2$ .

# Cross validation

- ▶ A very common approach for model selection is cross-validation.
- ▶ This is exactly the same procedure used in classical statistics.
- ▶ This operates under the assumption that the “true” model likely produces better out-of-sample predictions than competing models.
- ▶ Advantages: Simple, intuitive, and broadly applicable.
- ▶ Disadvantages: Slow because it requires several model fits and it is hard to say a difference is statistically significant.

# K-fold Cross validation

- 0 Split the data into  $K$  equally-sized groups
- 1 Set aside group  $k$  as test set and fit the model to the remaining  $K - 1$  groups
- 2 Make predictions for the test set  $k$  based on the model fit to the training data
- 3 Repeat steps 1 and 2 for  $k = 1, \dots, K$  giving a predicted value  $\hat{Y}_i$  for all  $n$  observations
- 4 Measure prediction accuracy, e.g.,

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

# Variants

- ▶ Usually  $K$  is either 5 or 10.
- ▶  $K = n$  is called “leave-one-out” cross-validation, which is great but slow.
- ▶ The predicted value  $\hat{Y}_i$  can be either the posterior predictive mean or median.
- ▶ Mean squared error (MSE) can be replaced with Mean absolute deviation

$$MAD = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|.$$

## Information criteria

- ▶ Several information criteria have been proposed that do not require fitting the model several times.
- ▶ Many are functions of the **deviance**, i.e., twice the negative log likelihood

$$D(\mathbf{Y}|\theta) = -2 \log[f(\mathbf{Y}|\theta)].$$

- ▶ Ideally, models will have small deviance.
- ▶ However, if a model is too complex it will have small deviance (overfitting).
- ▶ The Akaike information criteria has a complexity penalty

$$AIC = D(\mathbf{Y}|\hat{\theta}) + 2p,$$

where  $\hat{\theta}$  is the MLE.

- ▶ Model with smaller  $AIC$  are preferred.



# Bayesian information criteria (BIC)

- ▶ The Bayesian information criteria is similar

$$BIC = D(\mathbf{Y}|\hat{\theta}) + \log(n)p.$$

- ▶ This is motivated as an approximation to the log Bayes factor of the model compared to the null model.
- ▶ However, this is only an asymptotic (large  $n$ ) approximation.
- ▶ With large  $n$  the prior is irrelevant, and so this is not satisfying to a subjective Bayesian.

# Deviance information criteria (DIC)

- ▶ *DIC* is a popular Bayesian analog of *AIC* or *BIC*.
- ▶ Unlike CV, *DIC* requires only one model fit.
- ▶ Unlike BF, it can be applied to complex models.
- ▶ However, proceed with caution.
- ▶ *DIC* really only applies when the posterior is approximately normal, and will give misleading results when the posterior far from normality, e.g., bimodal.
- ▶ *DIC* is also criticized for selecting overly-complex models.

## Deviance information criteria (DIC)

- ▶ Let  $\bar{D} = E[D(Y|\theta)|\mathbf{Y}]$  be the posterior mean of the deviance.
- ▶ Denote  $\hat{\theta}$  as the posterior mean of  $\theta$ .
- ▶ The effective number of parameters is

$$p_D = \bar{D} - D(\mathbf{Y}|\hat{\theta}).$$

- ▶ DIC can be written like *AIC*,

$$DIC = \bar{D} + p_D = D(\mathbf{Y}|\hat{\theta}) + 2p_D.$$

- ▶ Models with small  $\bar{D}$  fit the data well.
- ▶ Models with small  $p_D$  are simple.
- ▶ We prefer models that are simple and fit well, so we select the model with smallest *DIC*.

# DIC

- ▶ The effective number of parameters is a useful measure of model complexity.
- ▶ Intuitively, if there are  $p$  parameters and we have uninformative priors then  $p_D \approx p$ .
- ▶ However,  $p_D \ll p$  if there are strong priors.
- ▶ For example, how many free degrees of freedom do we have with  $\theta \sim \text{Beta}(1, 1)$  versus  $\theta \sim \text{Beta}(1000, 1000)$ ?
- ▶ In some cases  $p_D$  has a nice closed form.

# DIC

- ▶ As with  $AIC$  or  $BIC$ , we compute  $DIC$  for all models under consideration and select the one with smallest  $DIC$ .
- ▶ Rule of thumb: a difference of  $DIC$  of less than 5 is not definitive and a difference greater than 10 is substantial.
- ▶ As with  $AIC$  or  $BIC$ , the actual value is meaningless, only differences are relevant.
- ▶  $DIC$  can only be used to compare models with the same likelihood.

# Watanabe-Akaike information criteria (WAIC)

- ▶ *WAIC* is an alternative to *DIC*.
- ▶ It is motivated as an approximation to leave-one-out CV.
- ▶ In the end *WAIC* has model-fit and model-complexity components.
- ▶ It is used the same as *DIC* with smaller *WAIC* being preferred.
- ▶ In practice the two often give similar results, but *WAIC* is arguably more theoretically justified.

# Watanabe-Akaike information criteria (WAIC)

- ▶ *WAIC* is written in terms of the posterior of the likelihood rather than parameters.

- ▶ Let  $m_i$  and  $v_i$  be the posterior mean and variance of

$$\log[f(Y_i|\boldsymbol{\theta})].$$

- ▶ The effective model size is  $p_W = \sum_{i=1}^n v_i$ .

- ▶ The criteria is

$$WAIC = -2 \sum_{i=1}^n m_i + 2p_W.$$

Thank you!